EE 121B: Principles of Semiconductor Device Design

Lecture 6:

Metal-Semiconductor Contacts (continued)

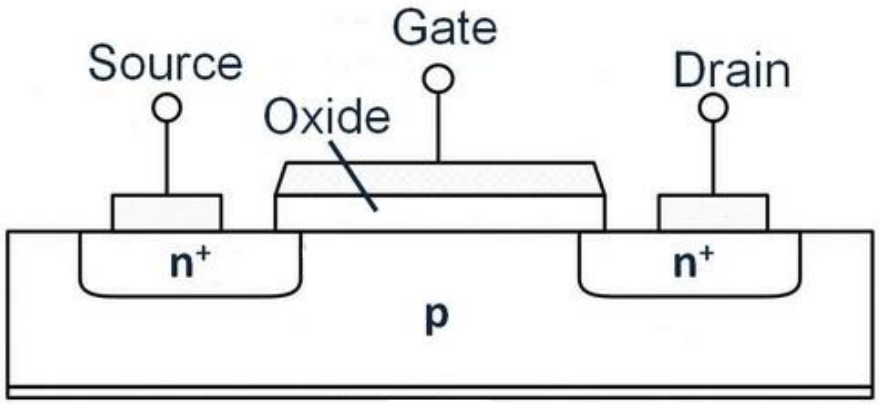
pn Junction Diodes

Sam Emaminejad

Course Overview

1. Semiconductor Properties
2. Metal-Semiconductor Contacts
3. P-N Junction
4. MOS Devices
5. Bipolar Junction Transistor

Metal-Oxide-Semiconductor (MOS)
Field-Effect Transistor (FET)



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Lecture 6

OUTLINE

- Metal-Semiconductor Contacts (cont'd)
- pn Junction Diodes
  - Electrostatics (step junction)

Reading: Pierret 14.2-14.3, 5; Hu 4.17-4.21, 4.1-4.2

Recall: $\Phi_M > \Phi_S$, n-type

The Depletion Approximation

The semiconductor is depleted of mobile carriers to a depth *W*

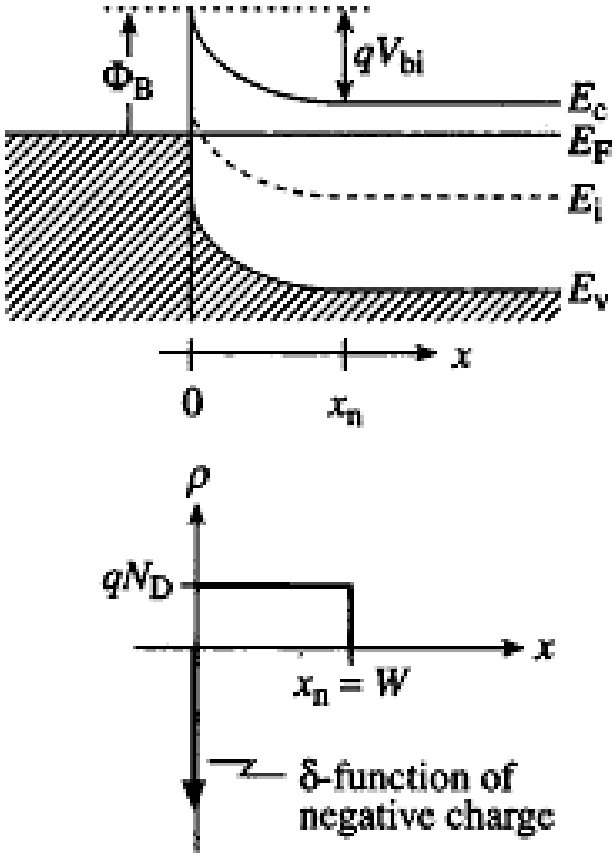
n and p each is much less than the net dopant concentration (N_D-N_A) within the depleted region

$\Rightarrow$ In the depleted region ($0 \leq x \leq W$):

$$\rho = q\,(N_D - N_A)$$

Beyond the depleted region ($x > W$):

$$\rho = 0$$



R.F. Pierret, *Semiconductor Fundamentals*, Fig. 14.4

Electrostatics

Poisson's equation
$\rho(x) \rightarrow \mathcal{E}(x) \rightarrow V(x)$

- Poisson's equation: $\dfrac{\partial \mathcal{E}}{\partial x} = \dfrac{\rho}{\varepsilon_s} \cong \dfrac{qN_D}{\varepsilon_s}$

$$\mathcal{E}(x) = \int \frac{qN_D}{\varepsilon_s} dx = Ax + B = \frac{qN_D}{\varepsilon_s} x + B$$

$$\text{slope} = A = \frac{d\mathcal{E}}{dx} = \frac{qN_D}{\varepsilon_s}$$

Boundary condition:

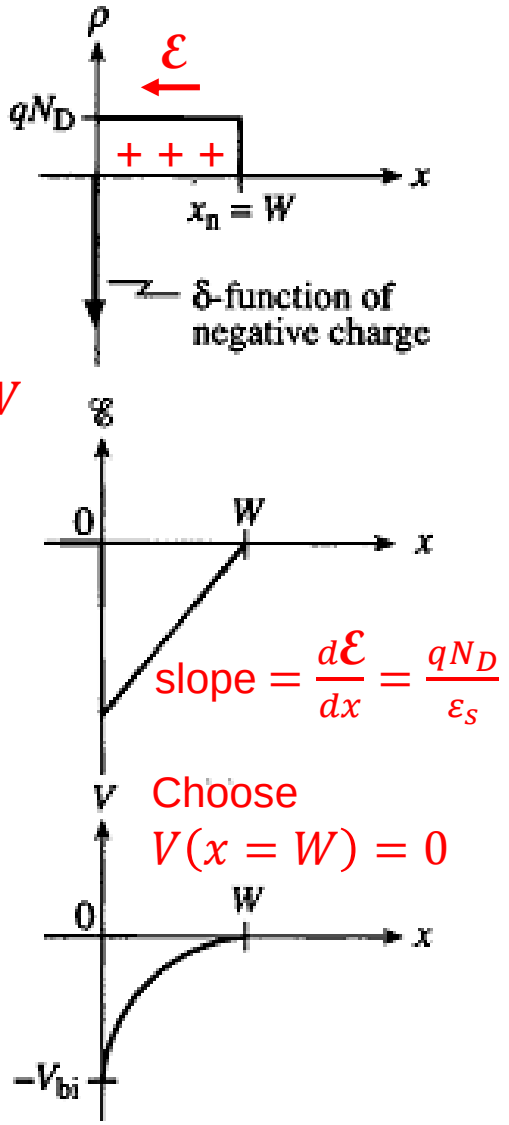
$$\mathcal{E}(W) = 0 \rightarrow \mathcal{E}(W) = \frac{qN_D}{\varepsilon_s} W + B = 0 \rightarrow B = -\frac{qN_D}{\varepsilon_s} W$$

- The solution is: $\mathcal{E}(x) = -\dfrac{qN_D}{\varepsilon_s}(W - x)$

$$\mathcal{E} = -\frac{dV}{dx}$$

$$V(x) = -\int \mathcal{E}(x')dx' = \int \frac{qN_D}{\varepsilon_s}(W - x)dx$$

$$= -\frac{qN_D}{2\varepsilon_s}(W - x)^2$$



R.F. Pierret, *Semiconductor Fundamentals*, Fig. 14.4

Depletion Width, *W*

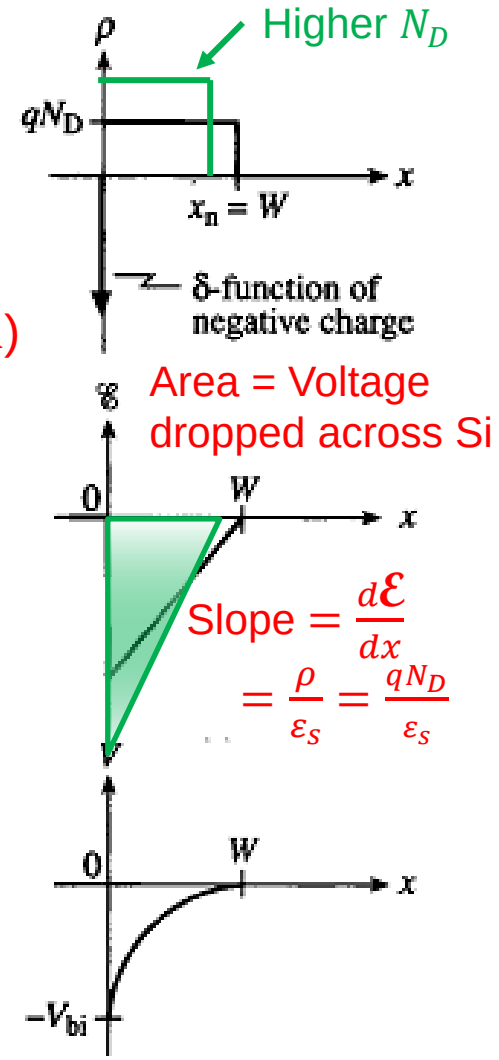
$$V(x) = \frac{-qN_D}{2K_S\varepsilon_0}(W - x)^2$$

11.7 for Si ($\varepsilon_{Si} = 10^{-12}$F/cm)

At $x = 0$, $V = -V_{bi}$

$$\Rightarrow \quad W = \sqrt{\frac{2\varepsilon_s V_{bi}}{qN_D}}$$

- ***W* decreases with increasing N_D**



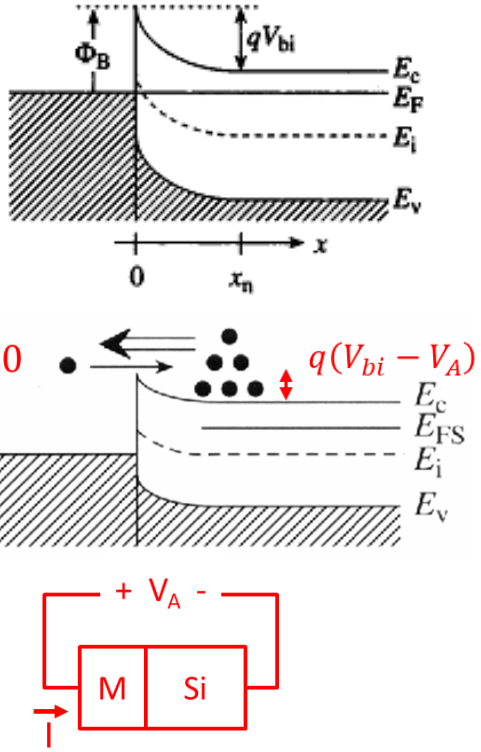
R.F. Pierret, *Semiconductor Fundamentals*, Fig. 14.4

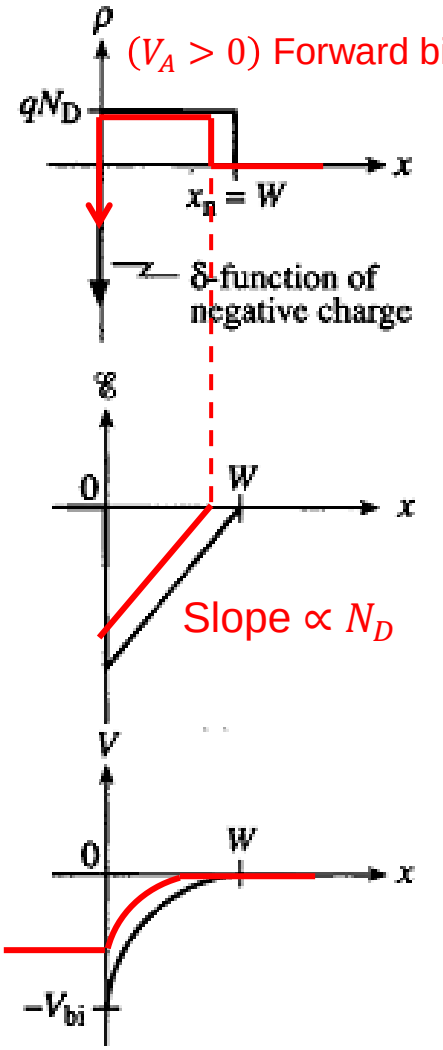
Depletion Width, W, for $V_A \neq 0$

$$V(x) = \frac{-qN_D}{2K_S\varepsilon_0}(W - x)^2$$

At $x = 0$, $V = -(V_{bi} - V_A)$

$$\Rightarrow \quad W = \sqrt{\frac{2\varepsilon_s(V_{bi} - V_A)}{qN_D}}$$

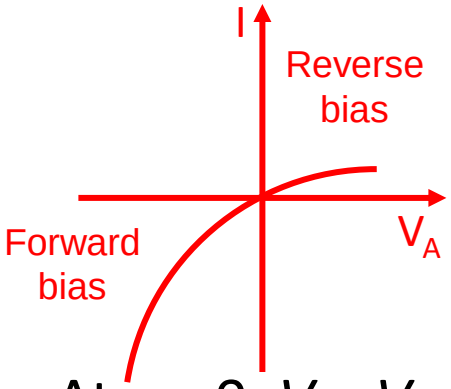




- ***W*** **increases with increasing** $-V_A$
- ***W*** **decreases with increasing** N_D

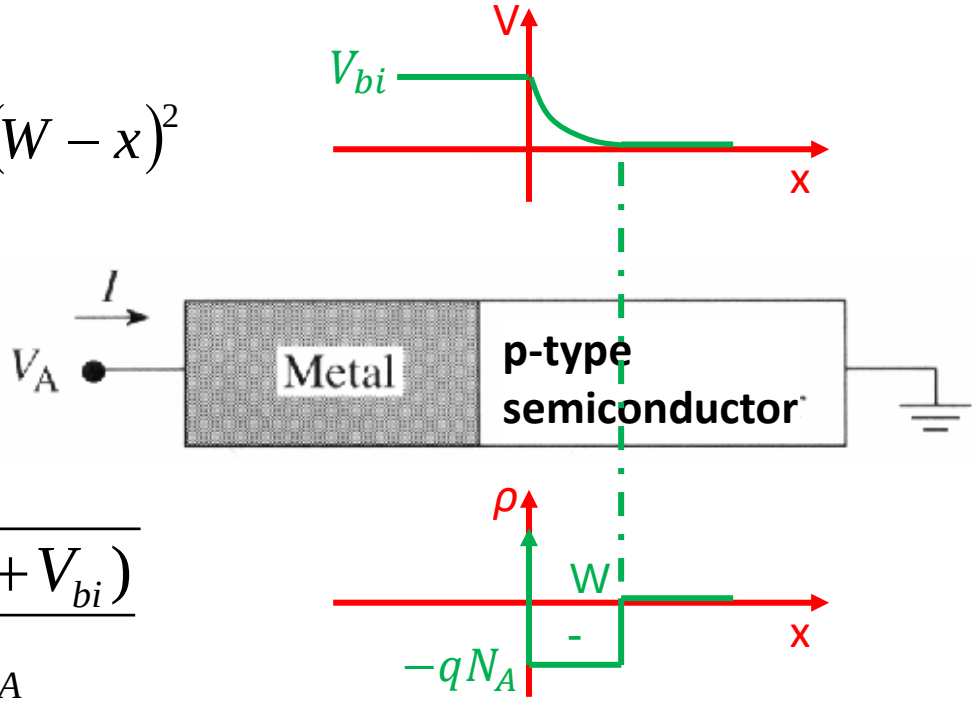
R.F. Pierret, *Semiconductor Fundamentals*, Fig. 14.4

W for p-type Semiconductor



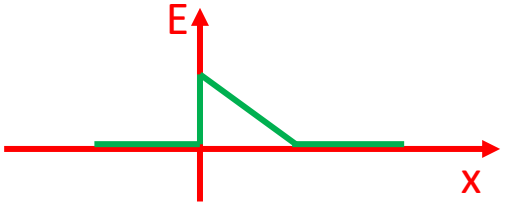
$$V(x) = \frac{qN_A}{2K_S\varepsilon_0}(W - x)^2$$

At $x = 0$, $V = V_{bi} + V_A$



$$\Rightarrow \quad W = \sqrt{\frac{2\varepsilon_s(V_A + V_{bi})}{qN_A}}$$

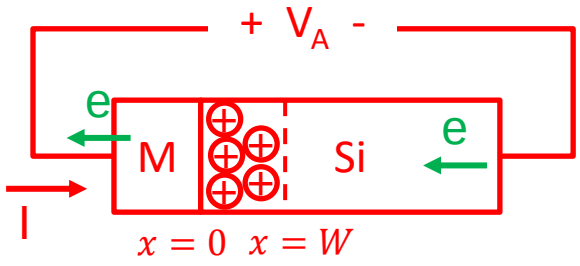
- ***W*** **increases with increasing** V_A
- ***W*** **decreases with increasing** N_A

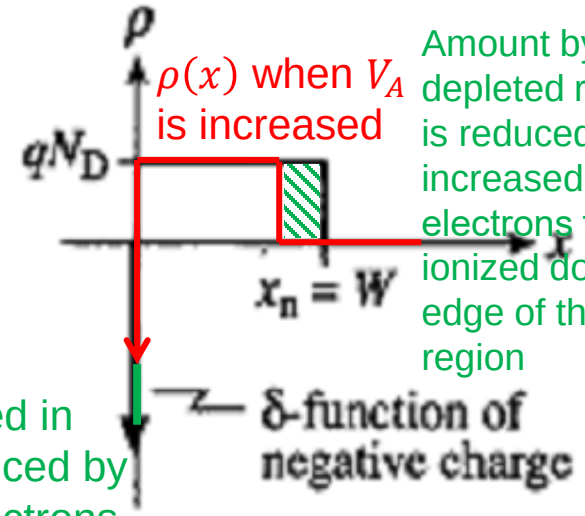


Charge Storage in a Schottky Diode

- Charge is "stored" on both sides of the M-S contact.
  - **The applied bias V_A modulates this charge.**

A change in V_A requires charge to be added/subtracted on each side of the contact





Amount by which the depleted region charge is reduced when V_A is increased, by adding electrons to "cover up" ionized donors at the edge of the depleted region

Charge stored in metal is reduced by removing electrons

R.F. Pierret, *Semiconductor Fundamentals*, Fig. 14.4

Small-Signal Capacitance

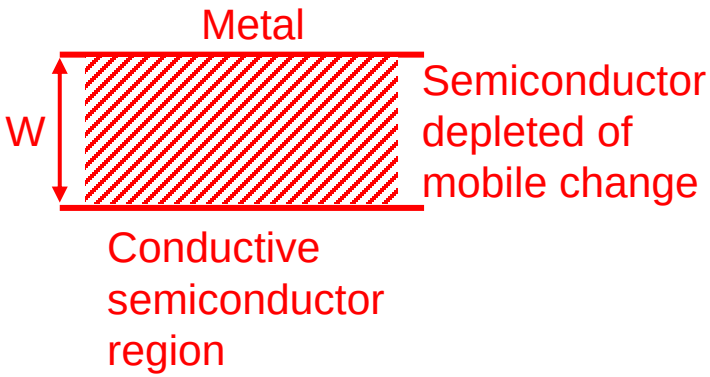
- If an a.c. voltage v_a is applied in series with the d.c. bias V_A, the charge stored in the Schottky contact will be modulated at the frequency of the a.c. voltage

$$\Delta Q = C\Delta V \rightarrow \frac{dQ}{dt} = C\frac{dv_a}{dt}$$

$\rightarrow$ displacement current will flow:

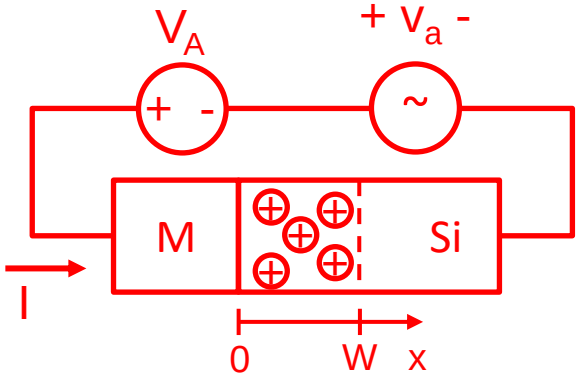
$$i = C\frac{dv_a}{dt}$$

Parallel-plate small-signed capacitance



$$C = A\frac{\varepsilon_s}{W}$$

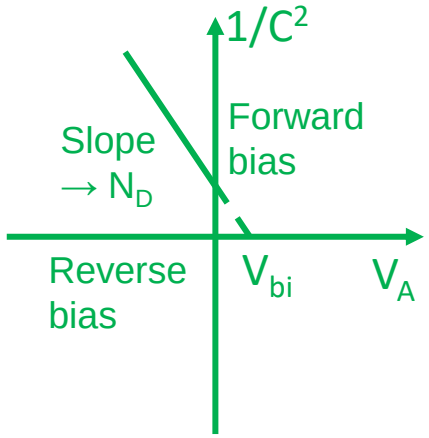
Junction area



Using *C-V* Data to Determine Φ_B

$$C = A\frac{\varepsilon_s}{W} = A\frac{\varepsilon_s}{\sqrt{\frac{2\varepsilon_s}{qN_D}(V_{bi} - V_A)}} = A\sqrt{\frac{qN_D\varepsilon_s}{2(V_{bi} - V_A)}}$$

$$\frac{1}{C^2} = \frac{2(V_{bi} - V_A)}{qN_D\varepsilon_s A^2}$$

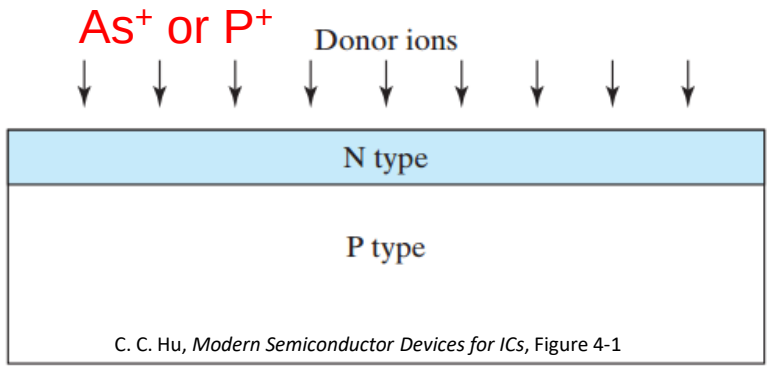


Once V_{bi} and N_D are known, Φ_{Bn} can be determined:

$$qV_{bi} = \Phi_{Bn} - (E_c - E_F)_{FB} = \Phi_{Bn} - kT\ln\frac{N_c}{N_D}$$

pn Junctions

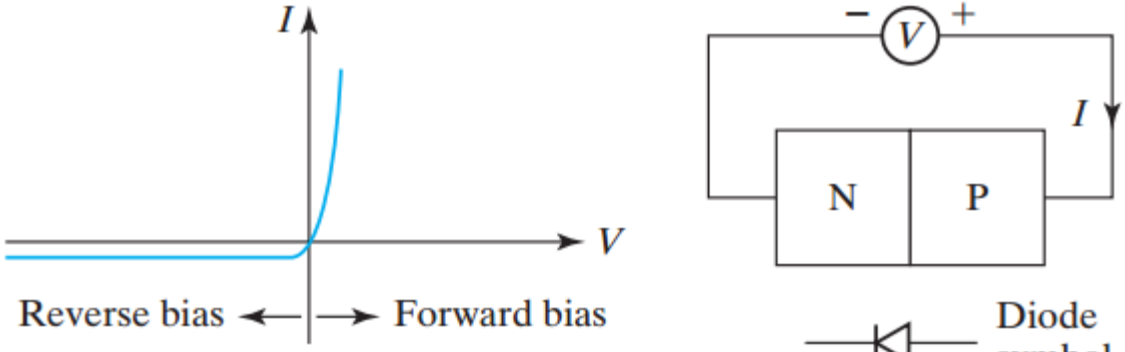
- A pn junction is typically fabricated by implanting or diffusing donor atoms into a p-type substrate to form an n-type layer:



C. C. Hu, *Modern Semiconductor Devices for ICs*, Figure 4-1

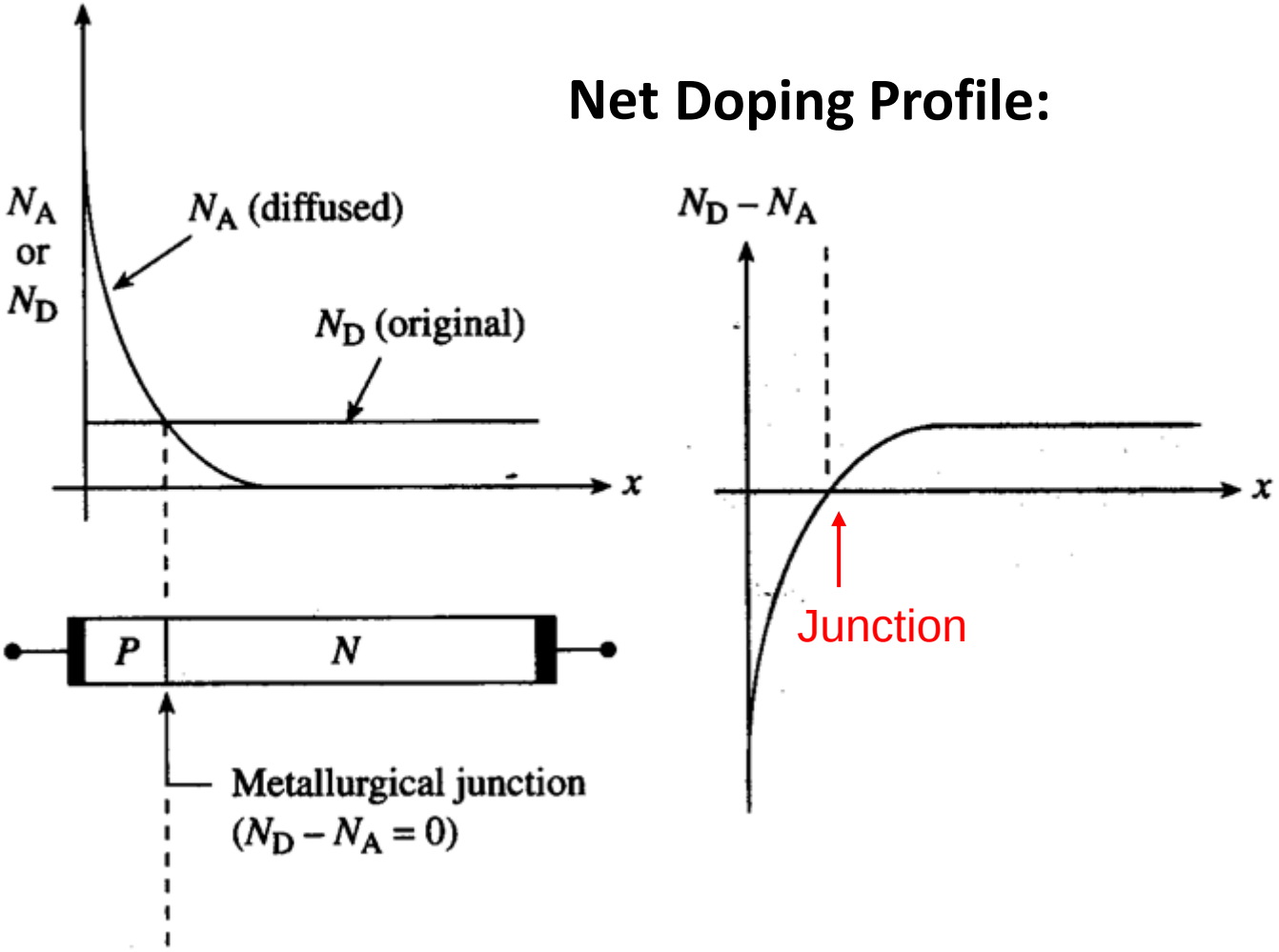
Wafer surface is neutralized by an electron gun as ions are implemented

- A pn junction has a rectifying current-*vs*.-voltage characteristic:



C. C. Hu, *Modern Semiconductor Devices for ICs*, Figure 4-2

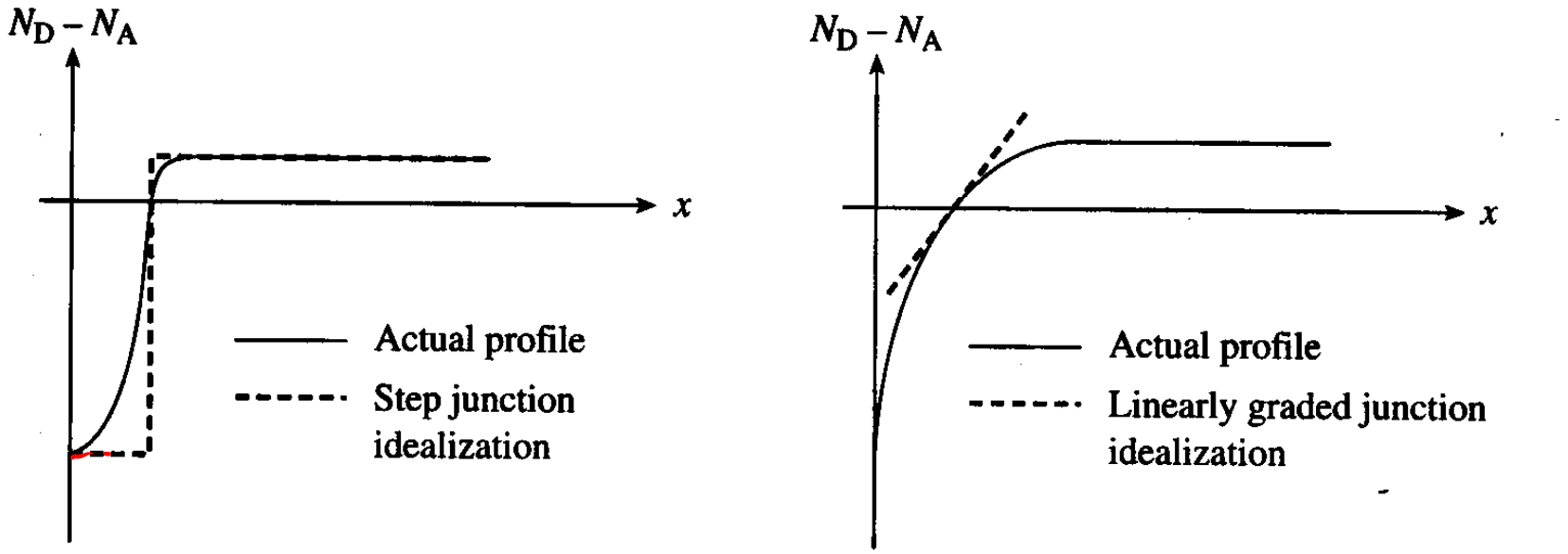
Terminology



R.F. Pierret, *Semiconductor Fundamentals*, Figure 5.1

Idealized pn Junctions

R.F. Pierret, *Semiconductor Fundamentals*, Figure 5.2



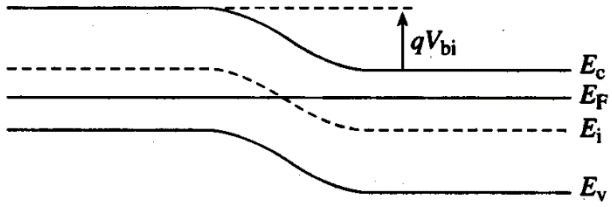
- **In the analysis going forward, we will consider only the net dopant concentration on each side of the pn junction:**

  $N_A \equiv$ net acceptor doping on the p side: $(N_A\text{-}N_D)_{\text{p-side}}$

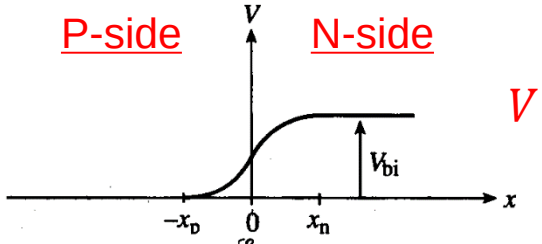
  $N_D \equiv$ net donor doping on the n side: $(N_D\text{-}N_A)_{\text{n-side}}$

Electrostatics (Step Junction)

Band diagram:

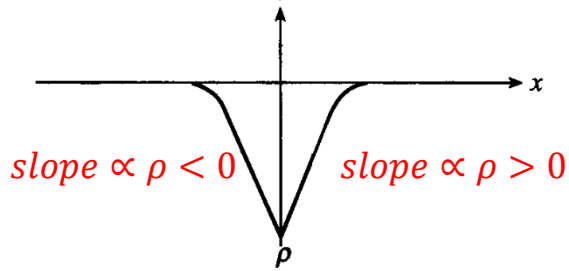


Electrostatic potential:



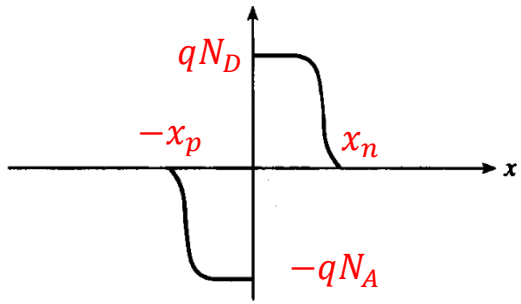
$$V = \frac{1}{q}(E_{ref} - E_c)$$

Electric field: $\mathcal{E} = -\frac{dV}{dx}$



$$\frac{d\mathcal{E}}{dx} = \frac{\rho}{\varepsilon_s}$$

Charge density:



Depletion width
$W = x_n + x_p$

R.F. Pierret, *Semiconductor Fundamentals*, Figure 5.4

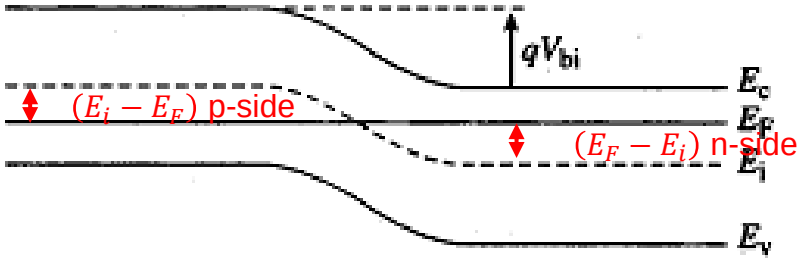
"Game Plan" to obtain $\rho(x)$, $\mathcal{E}(x)$, $V(x)$

1. Find the built-in potential V_{bi}
2. Use the depletion approximation $\rightarrow \rho(x)$
   (depletion widths x_p, x_n unknown)
3. Integrate ρ (x) to find $\mathcal{E}$(x)
   Apply boundary conditions $\mathcal{E}(-x_p)=0$, $\mathcal{E}(x_n)=0$
4. Integrate $\mathcal{E}$(x) to obtain V(x)
   Apply boundary conditions $V(-x_p)=0$, $V(x_n)=V_{bi}$
5. For $\mathcal{E}$(x) to be continuous at x=0, $N_A x_p = N_D x_n$
   Solve for x_p, x_n

Built-In Potential V_{bi}

$$qV_{bi} = \Phi_{S\,p-side} - \Phi_{S\,n-side} = (E_i - E_F)_{p-side} + (E_F - E_i)_{n-side}$$

Difference between work functions on p-side vs. n-side



R.F. Pierret, *Semiconductor Fundamentals*, Figure 5.4a

For **non-degenerately doped** material:

$$p = n_i e^{(E_i - E_F)/kT} \qquad n = n_i e^{(E_F - E_i)/kT}$$

$$(E_i - E_F)_{p-side} = kT \ln\left(\frac{p}{n_i}\right) = kT \ln\left(\frac{N_A}{n_i}\right)$$

$$(E_F - E_i)_{n-side} = kT \ln\left(\frac{n}{n_i}\right) = kT \ln\left(\frac{N_D}{n_i}\right)$$

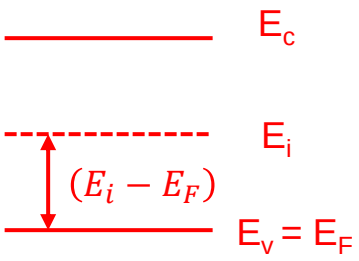
$$qV_{bi} = kT ln\left(\frac{N_A}{n_i}\right) + kT ln\left(\frac{N_D}{n_i}\right) = kT ln\left(\frac{N_A N_D}{n_i^2}\right)$$

What if one side is degenerately doped?

$$qV_{bi} = (E_i - E_F)_{p-side} + (E_F - E_i)_{n-side}$$

p⁺n junction

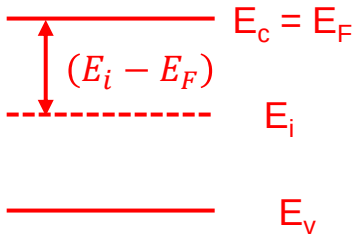
For p⁺-Si, $(E_i - E_F) = \frac{E_G}{2}$



$$qV_{bi} = \frac{E_G}{2} + kTln\left(\frac{N_D}{n_i}\right)$$

n⁺p junction

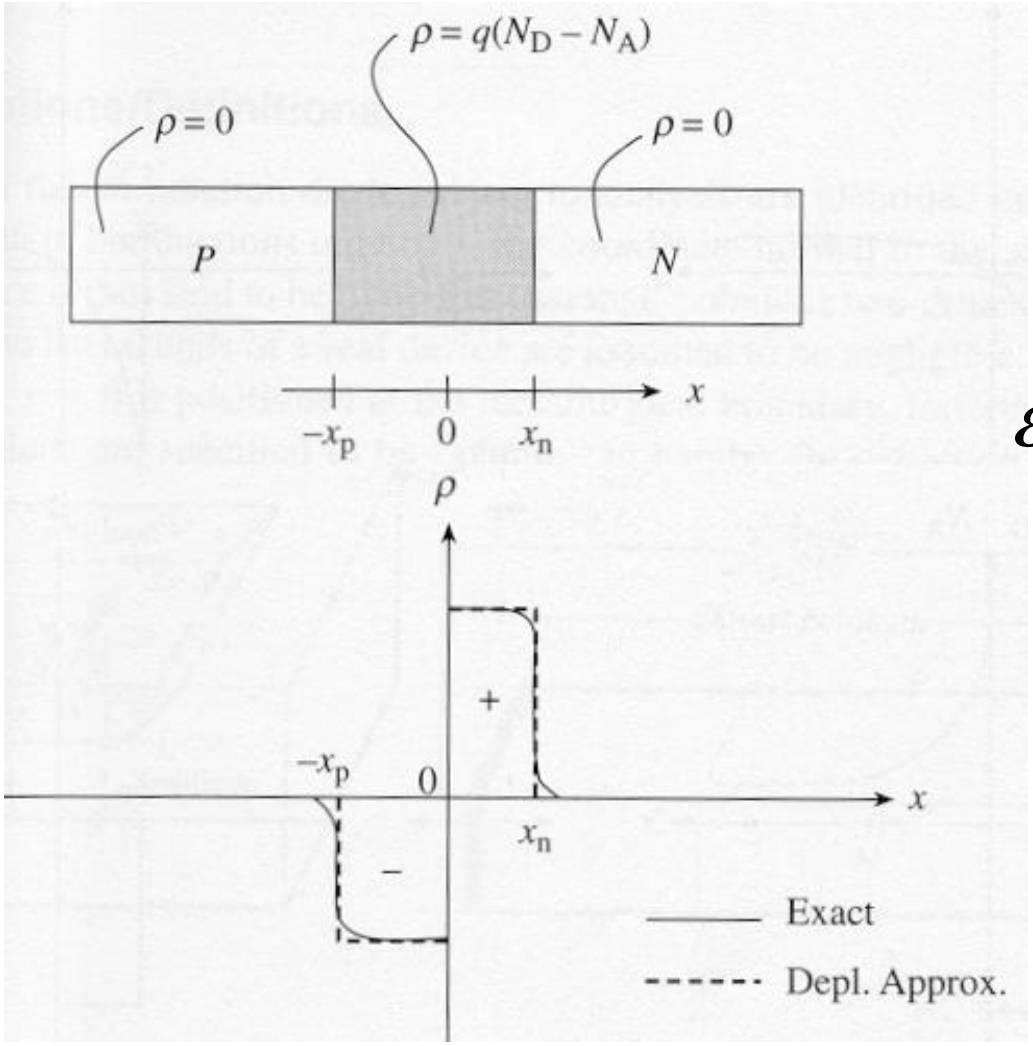
For n⁺-Si, $(E_F - E_i) = \frac{E_G}{2}$



$$qV_{bi} = \frac{E_G}{2} + kTln\left(\frac{N_A}{n_i}\right)$$

The Depletion Approximation

R.F. Pierret, *Semiconductor Fundamentals*, Figure 5.6



In the depletion region on the p side, $\rho = -qN_A$

Poisson's Eq'n: $\frac{d\mathcal{E}}{dx} = \frac{-qN_A}{\varepsilon_s}$

$$\mathcal{E}(x) = -\frac{qN_A}{\varepsilon_s}x + C_1 = -\frac{qN_A}{\varepsilon_s}\left(x + x_p\right)$$

$@x = -x_p, \mathcal{E} = 0$

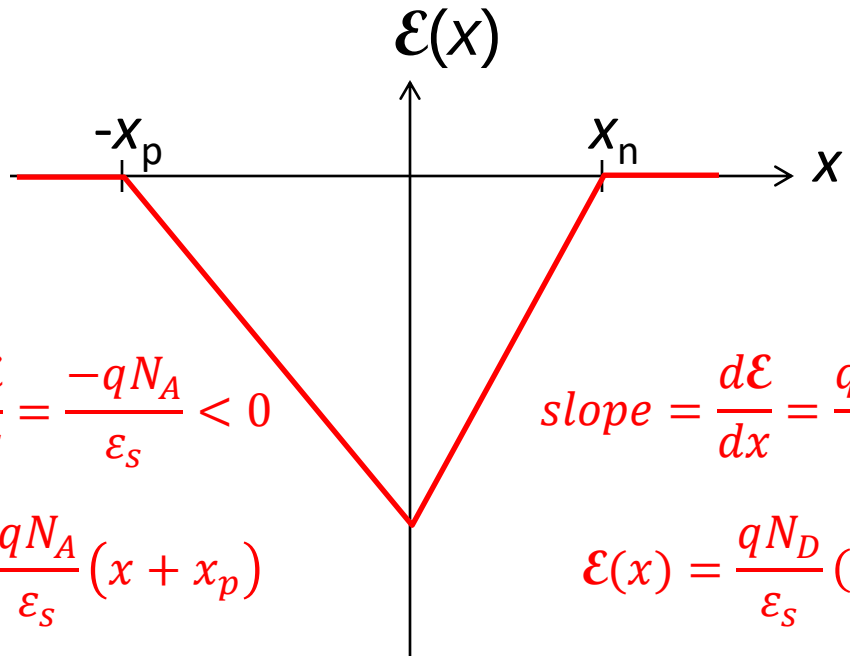
In the depletion region on the n side, $\rho = qN_D$

Poisson's Eq'n: $\frac{d\mathcal{E}}{dx} = \frac{qN_D}{\varepsilon_s}$

$$\mathcal{E}(x) = \frac{qN_D}{\varepsilon_s}x + C_1 = \frac{qN_D}{\varepsilon_s}\left(x - x_n\right)$$

$@x = x_n, \mathcal{E} = 0$

Electric Field Distribution



$$slope = \frac{d\mathcal{E}}{dx} = \frac{-qN_A}{\varepsilon_s} < 0$$

$$\mathcal{E}(x) = -\frac{qN_A}{\varepsilon_s}(x + x_p)$$

$$slope = \frac{d\mathcal{E}}{dx} = \frac{qN_D}{\varepsilon_s} > 0$$

$$\mathcal{E}(x) = \frac{qN_D}{\varepsilon_s}(x - x_n)$$

$$\mathcal{E}(x = 0-) = -\frac{qN_A}{\varepsilon_s}x_p$$

$$\mathcal{E}(x = 0+) = -\frac{qN_D}{\varepsilon_s}x_n$$

The electric field is continuous at $x = 0$

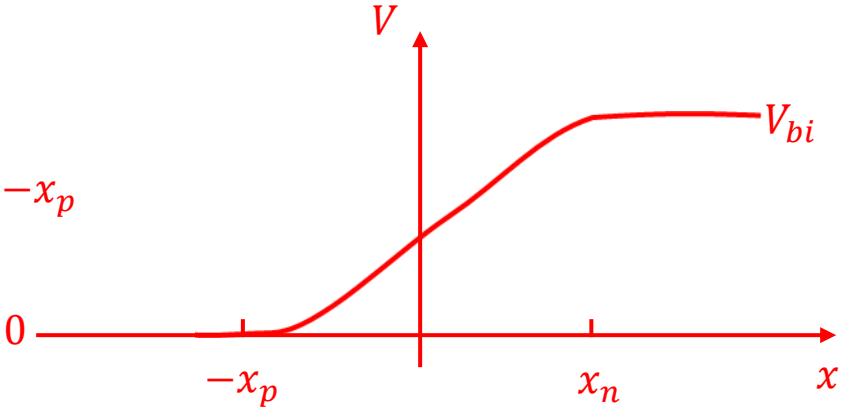
$$\rightarrow N_A x_p = N_D x_n$$

Electrostatic Potential Distribution

On the p side: $\mathcal{E}(x) = -\frac{qN_A}{\varepsilon_s}(x + x_p)$ $\mathcal{E} = -\frac{dV}{dx}$

$$V(x) = \frac{qN_A}{2\varepsilon_s}(x + x_p)^2 + \cancel{D_1}^{\,0}$$

If $V = 0$, @ $x = -x_p$



Choose $V(-x_p)$ to be 0

$\Rightarrow$ **$V(x_n) = V_{bi}$**

On the n side: $\mathcal{E}(x) = \frac{qN_D}{\varepsilon_s}(x - x_n)$ $\mathcal{E} = -\frac{dV}{dx}$

$$V(x) = -\frac{qN_D}{2\varepsilon_s}(x_n - x)^2 + \cancel{D_2}^{\,V_{bi}} = V_{bi} - \frac{qN_D}{2\varepsilon_s}(x_n - x)^2$$

$V = V_{bi}$, @ $x = x_n$

Derivation of Depletion Width

- At $x = 0$, expressions for p side and n side must be equal:

$$V(x = 0^-) = \frac{qN_A}{2\varepsilon_s}(0 + x_p)^2 \qquad V(x = 0^+) = V_{bi} - \frac{qN_D}{2\varepsilon_s}(x_n - 0)^2$$

$$\frac{qN_A}{2\varepsilon_s}x_p^2 = V_{bi} - \frac{qN_D}{2\varepsilon_s}x_n^2$$

- We also know that $N_A x_p = N_D x_n$

$$\rightarrow x_p = \frac{N_D}{N_A}x_n \ \ or \ x_n = \frac{N_A}{N_D}x_p$$

Depletion Width

- Eliminating x_p, we have:

$$x_n = \sqrt{\frac{2\varepsilon_s V_{bi}}{q}\left(\frac{N_A}{N_D(N_A + N_D)}\right)}$$

- Eliminating x_n, we have:

$$x_p = \sqrt{\frac{2\varepsilon_s V_{bi}}{q}\left(\frac{N_D}{N_A(N_A + N_D)}\right)}$$

- Summing, we have:

$$x_n + x_p = W = \sqrt{\frac{2\varepsilon_s V_{bi}}{q}\left(\frac{1}{N_A} + \frac{1}{N_D}\right)}$$

Depletion Width in a One-Sided Junction

$$x_n + x_p = W = \sqrt{\frac{2\varepsilon_s V_{bi}}{q}\left(\frac{1}{N_A} + \frac{1}{N_D}\right)}$$

If $N_A >> N_D$ as in a p⁺n junction: $\frac{1}{N_A} + \frac{1}{N_D} \cong \frac{1}{N_D}$

$$W = \sqrt{\frac{2\varepsilon_s V_{bi}}{qN_D}} \approx x_n$$

$$x_p = x_n N_D / N_A \cong 0$$

What about a n⁺p junction? $\frac{1}{N_A} + \frac{1}{N_D} \cong \frac{1}{N_A}$ $W = \sqrt{\frac{2\varepsilon_s V_{bi}}{qN_A}} \cong x_p$

$N_D \gg N_A$

$$W = \sqrt{2\varepsilon_s V_{bi} / qN} \quad \text{where} \quad \frac{1}{N} = \frac{1}{N_D} + \frac{1}{N_A} \approx \frac{1}{\textit{lighter dopant density}}$$

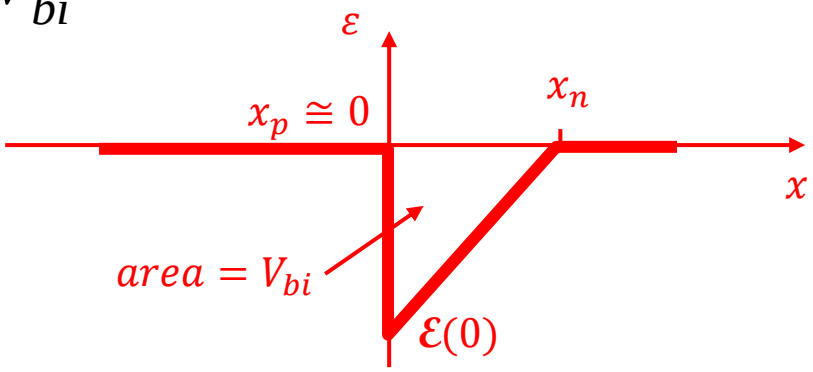
Peak $\mathcal{E}$-Field in a One-Sided Junction

$$\left| \int \mathcal{E}\ dx \right| = \frac{1}{2}\left|\mathcal{E}(0)\right| W = V_{bi}$$

$$W \cong \sqrt{\frac{2\varepsilon_s}{qN} V_{bi}}$$

$$\mathcal{E}(0) = \frac{2V_{bi}}{W} \cong \sqrt{\frac{2qNV_{bi}}{\varepsilon_s}}$$

Example: p⁺n junction



$V(x)$ in a One-Sided Junction

p side

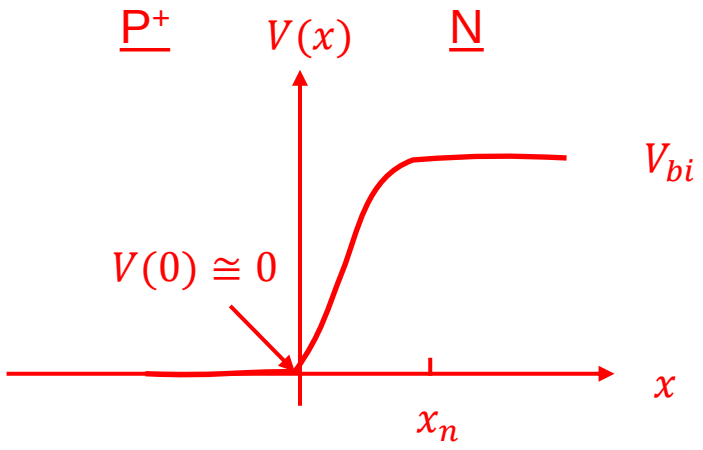
$$V(x) = \frac{qN_A}{2\varepsilon_s}(x + x_p)^2$$

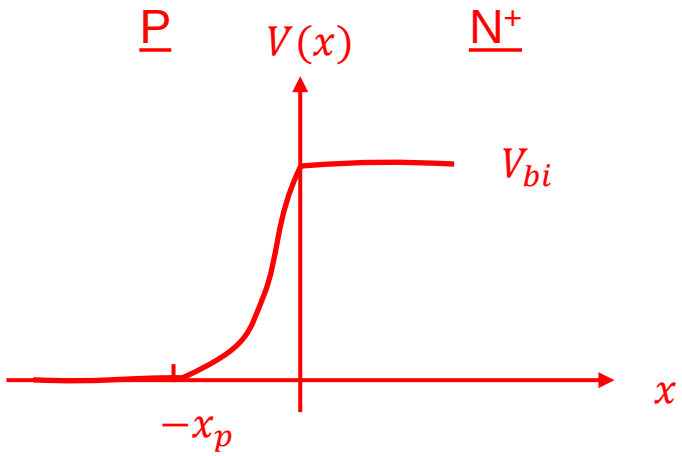
n side

$$V(x) = V_{bi} - \frac{qN_D}{2\varepsilon_s}(x_n - x)^2$$

$$V(0) = \frac{N_D}{N_A + N_D} V_{bi}$$

Most of voltage is dropped across the more lightly doped side





$\varepsilon_s \cong 10^{-12} F/cm$

Example: One-Sided pn Junction

A p^+n junction has $N_A=10^{20}\,cm^{-3}$ and $N_D=10^{17} cm^{-3}$.
Find (a) V_{bi} (b) W (c) x_n and (d) x_p.

$$V_{bi} = \frac{E_G}{2q} + \frac{kT}{q}\ln\frac{N_D}{n_i} \qquad \cong 0.56 + 0.42 = 0.98\,V \sim 1\,V$$

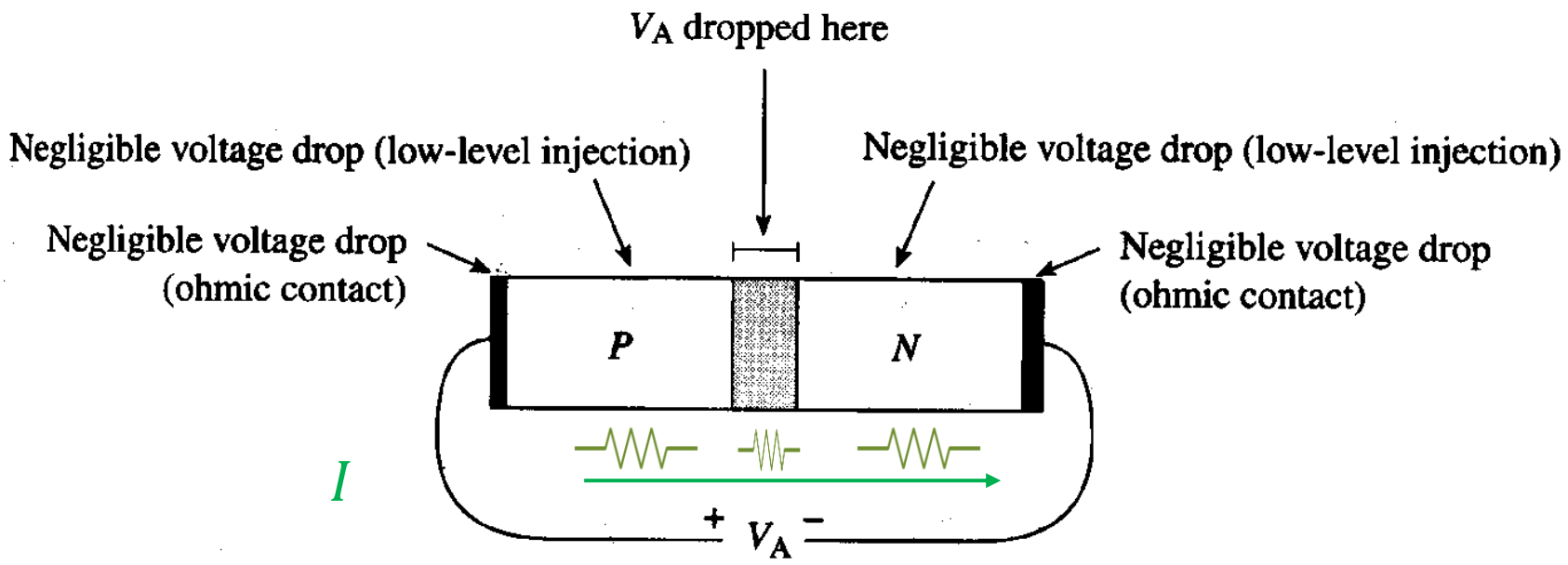
$$W \approx \sqrt{\frac{2\varepsilon_s V_{bi}}{qN_D}} \qquad = \left[\frac{2\times 10^{-12}}{1.6\times 10^{-19}\times 10^{17}}\right]^{1/2} \cong 10^{-5} cm = 0.1\mu m$$

$$x_n \cong W \qquad \cong 0.1\mu m$$

$$x_p = x_n\, N_D/N_A \qquad \cong 1\text{Å} \cong 0$$

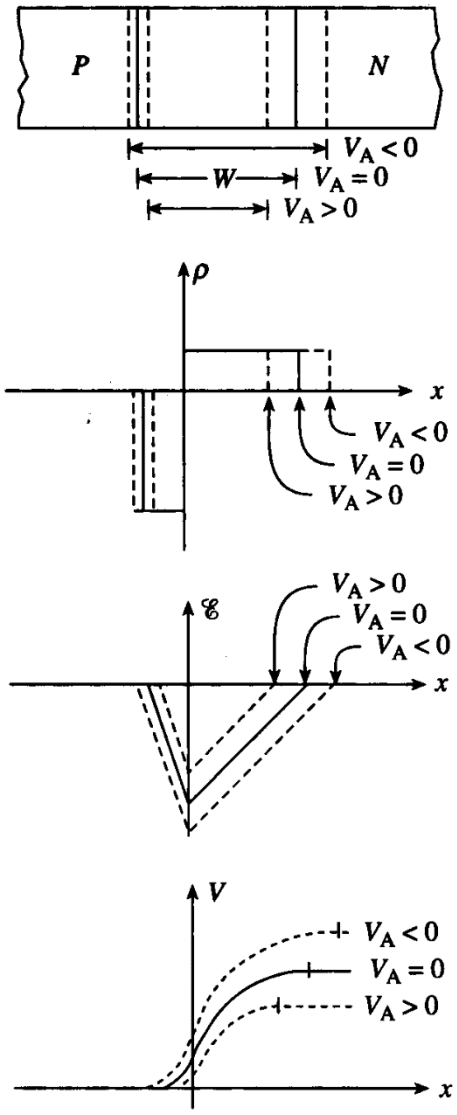
Voltage Drop across a pn Junction

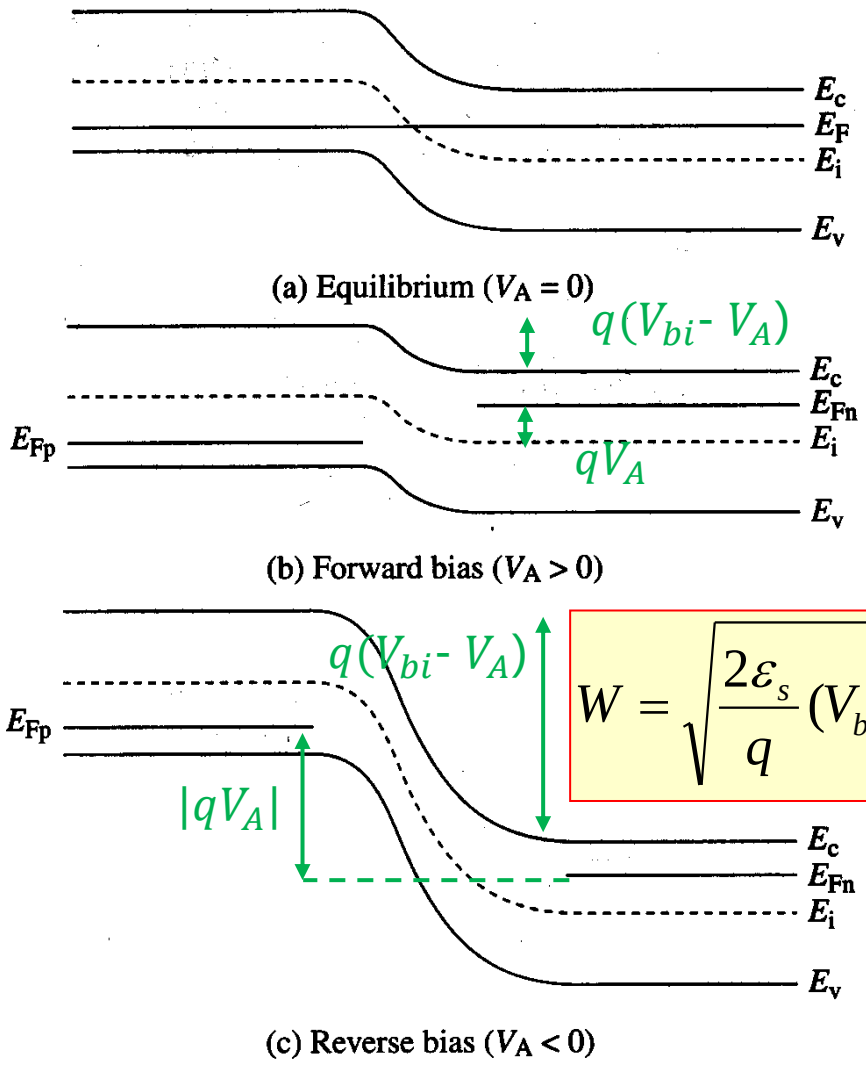
R.F. Pierret, *Semiconductor Fundamentals*, Figure 5.10



Note that V_A should be significantly smaller than V_{bi} in order for low-level injection conditions to prevail in the quasi-neutral regions.

Effect of Applied Voltage





$Note: qV_{bi} < E_G$

$$W = \sqrt{\frac{2\varepsilon_s}{q}(V_{bi} - V_A)\left(\frac{1}{N_A} + \frac{1}{N_D}\right)}$$

R.F. Pierret, *Semiconductor Fundamentals*, Figure 5.11

Summary

- **For a non-degenerately-doped pn junction:**

Built-in potential $$V_{bi} = \frac{kT}{q} \ln \frac{N_D N_A}{n_i^2}$$

Depletion width $$W = x_n + x_p = \sqrt{\frac{2\varepsilon_s \left(V_{bi} - V_A\right)}{q} \left(\frac{1}{N_A} + \frac{1}{N_D} \right)}$$

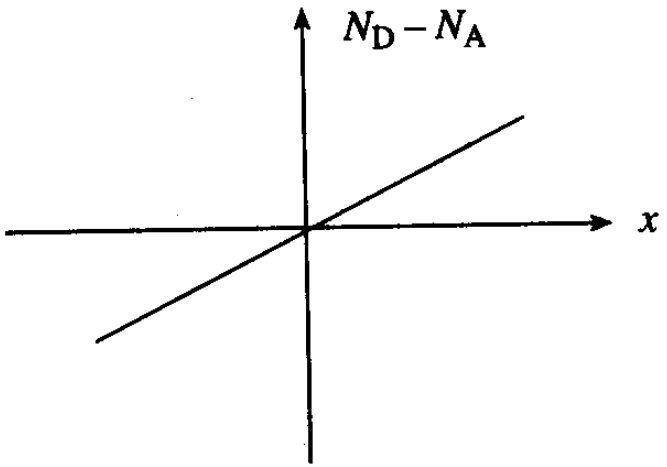
$$x_n = \frac{N_A}{N_A + N_D} W \qquad x_p = \frac{N_D}{N_A + N_D} W$$

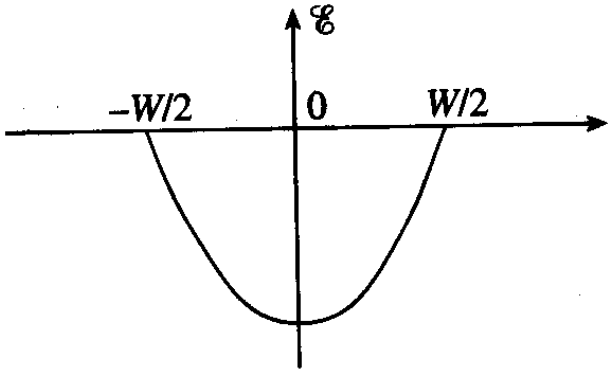
- **For a one-sided junction:**

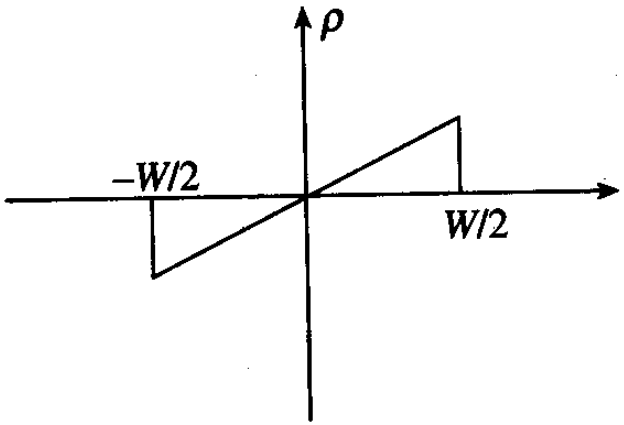
Built-in potential $$V_{bi} = \frac{E_G}{2} + \frac{kT}{q} \ln \frac{N}{n_i}$$

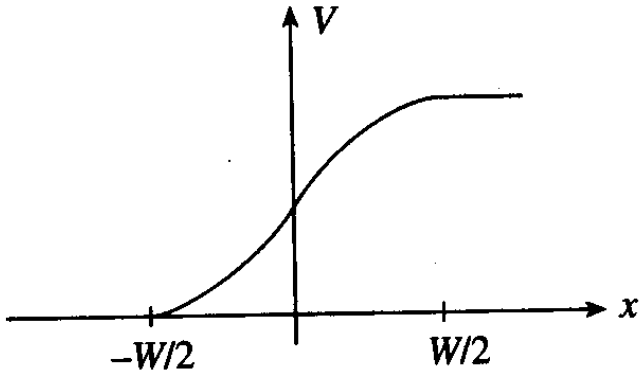
Depletion width $$W = \sqrt{\frac{2\varepsilon_s \left(V_{bi} - V_A\right)}{qN}}$$

Linearly Graded pn Junction









Acknowledgement

- Modern Semiconductor Devices for Integrated Circuits by C. Hu (Prentice Hall, 2009)
- Semiconductor Device Fundamentals by R. F. Pierret (Addison Wesley, 1996)
- Professor Tsu-Jae King Liu EE130/230A notes